

Kai Zhu

PH. D. STUDENT · ZHEJIANG UNIVERSITY

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Summary

I am a Ph.D. student in Pharmaceutics at Zhejiang University, advised by Prof. Tingjun Hou. I am also a visiting student at the Italian Institute of Technology, working with Luigi Bonati and Prof. Michele Parrinello. My research focuses on developing machine learning-based enhanced sampling methods and their applications to atomic systems.

Education

College of Pharmaceutical Sciences, Zhejiang University

PH.D. IN MEDICINE CHEMISTRY

Hangzhou, China

Sep. 2023 – Jun. 2028

College of Pharmaceutical Sciences, Wenzhou Medical University

B.S. IN PHARMACEUTICAL SCIENCE

Wenzhou, China

Sep. 2019 – Jun. 2023

Research Experience

Italian Institute of Technology

RESEARCH INTERN, ADVISED BY PHD. LUIGI BONATI AND PROF. MICHELE PARRINELLO

Genova, Italy

Dec. 2025 – Dec. 2027

Research Topics: machine learning, enhanced sampling, atomic simulation

Zhejiang University

PHD STUDENT, ADVISED BY PROF. TINGJUN HOU AND PROF. DAN LI

Hangzhou, China

Jun. 2023 – Jun. 2025

Research Topics: machine learning, enhanced sampling, intrinsically disordered proteins

Honors & Awards

DOMESTIC

2025 Qiu Shi Flying Eagle Scholarship, Zhejiang University

Hangzhou, China

Selected Publications

† Co-first author; * Corresponding author;

- 2026 **Contrastive Learning of Dynamical Representations for Enhanced Molecular Sampling** [4]
K. Zhu†, J. Zhang†, P. Novelli, T. Hou*, L. Bonati*. **arXiv** arXiv:2606.15495 (2026).
- 2026 **Targeting the Intrinsically Disordered AR-NTD through a Machine Learning-Based Enhanced Sampling Workflow** [3]
K. Zhu†, H. Wang†, J. Zhang†, R. Hu, L. Jiang, H. Zhang, Y. Kang, T. Hou*, D. Li*. **Nat. Commun.** (2026).
- 2026 **A Scalable and Quantum-Accurate Foundation Model for Biomolecular Force Field via Linearly Tensorized Quadrangle Attention** [2]
Q. Su†, **K. Zhu†**, Q. Gou†, ..., J. Wang*, C. Hsieh*, T. Hou*. **Nature Communications** 17, 3639 (2026).
- 2025 **Enhanced Sampling in the Age of Machine Learning: Algorithms and Applications** [1]
K. Zhu†, E. Trizio†, J. Zhang, R. Hu, L. Jiang, T. Hou*, L. Bonati*. **Chemical Reviews** 126 (1), 671-713.

Skills

Programming	Python, C++, C, $\LaTeX$, Linux
Frameworks	PyTorch, TensorFlow, NumPy, Git, Anaconda
Atomic simulation	GROMACS, PLUMED, AMBER, OpenMM, ASE, LAMMPS, CP2K